RBI Grade B 2022 Quant - Recollected Questions

Quantitative Aptitude

Direction (1-3): In the following questions three equations are given in variables x. Third equation is equal to the sum of the first two equations. You have to solve the questions based on given information.

I. $ax^2 + bx + 4 = 0$

II. $ax^2 + 3x + c = 0$

III. $2x^2 + 7x + \sqrt{49 - 13} = 0$

Note: Equation III = Equation I + Equation II

Q1. What is the value of (b + c)?

A. 9
B. 7
C. 15
D. 6
E. 10

Q2. What is the product of the roots of equation III.

A. 3
B. 16
C. 22
D. 14
E. 8

Q3. What is the square of the larger root of equation II?

A. 1
B. 25
C. 16
D. 9
E. 4

Q4. Lateral surface area of a cube is 420 cm^2 less than the lateral surface area of a cylinder. Height of cylinder is 2R cm and radius is R cm. If the side of cube is equal to the radius of cylinder, find the approximate area of a circle whose radius is (R+3) cm.

A. 628 cm^2
B. 314 cm^2
C. 289 cm^2
D. 356 cm^2
E. 414 cm^2

Direction (5-6): Each question below contains a statement followed by Quantity I and Quantity II. You have to study the information along with the question and compare the value derived from Quantity I and Quantity II, then answer:

Q5. Quantity I: Value of 'p' such that $\left(\sqrt{a^p}\right)^2 = \frac{a^4}{\sqrt{a^{16}}}$

Quantity II: 1

A. Quantity I > Quantity II
B. Quantity I < Quantity II
C. Quantity I ≥ Quantity II
D. Quantity I ≤ Quantity II
E. Quantity I = Quantity II or no relation

Q6. Quantity I: $\frac{a^3 - b^3}{a-b} - 3ab$

Quantity II: $1 - \frac{1}{a} - \frac{1}{b}$

Note: a > 1 > b > 0

A. Quantity I > Quantity II
B. Quantity I < Quantity II
C. Quantity I ≥ Quantity II
D. Quantity I ≤ Quantity II
E. Quantity I = Quantity II or no relation

Direction: In the following question two equations are given in variables x and y. You have to solve these equations and determine the relation between x and y

Q7. I. $x^2 + 7x + 12 = 0$

II. $y^2 + 8y + 15 = 0$

A. x > y
B. x < y
C. x ≥ y
D. x ≤ y
E. x = y or no relation between x and y

Direction (8-12): Information regarding five different shops A, B, C, D, and E is given below. Each shop sells blue and red pens. Some data is missing which you have to calculate according to the question.

Shop	Number of Pens sold (Blue + Red)	Total Selling Price of pens
A	55	--
B	62	1080
C	--	--
D	51	--
E	--	1450

Selling Price of Blue pen = Rs. 20 and Red = Rs. 15 respectively.

Q8. Find the ratio of the number of blue pens to number of red pens sold by shop B.

A. 16:15
B. 11:15
C. 15:16
D. 12:13
E. 13:12

Q9. Ratio between number of blue pens sold by shop A and number of red pens sold by shop A is 3:2. If total selling price obtained by shop C is Rs 50 less than that obtained by shop A and number of blue pens sold by shop C is 5 more than red pens sold then, number of red pens sold by shop C is?

A. 24
B. 28
C. 21
D. 31
E. 34

Q10. If the ratio of total selling price of shop B and shop D is in the ratio 108:89 respectively. Then find the difference between the number of blue and red pens sold by shop D.

A. 6
B. 11
C. 9
D. 1
E. 15

Q11. If the sum of total selling priced of shop D and E is Rs. 2340 and the average of the red pens sold by these two shops is 30. Then find the number of blue pens sold by shop E.

A. 61
B. 59
C. 41
D. 57
E. 47

Q12. Find the average number of red pens sold by shop D, B and shop E together if the total selling price of shop D is $8\frac{1}{3}$% less than that of shop B and the ratio between blue pen sold by shop B and E is 30:47?

A. 27
B. 30
C. 24
D. 29
E. 22

Direction (13-15): A series is given below where the first term is marked as (a), the second as (b), third as (c) and so on.
Series – 100, 95, 105, 88, 114, ?
(a) (b) (c) (d) (e) (f)

Q13. What is the value of '(f)'?

A. 67
B. 87
C. 77
D. 74
E. 81

Q14. If a new series is made following the same pattern of the above series, where (a) is the third term, (b) is the fourth term, (c) is the fifth term and so on. Then find the second term of such a series.

A. 99
B. 98
C. 102
D. 111
E. None of the above

Q15. What minimum number should be added to the 'f+20' to get to the nearest perfect square.

A. 4
B. 1
C. 7
D. 3
E. 6

Direction (16-20): Read the following information carefully and answer the questions based on it.

In a locality there are four buildings namely A, B, C, and D. Each building has a certain number of floors. Each floor has a certain number of flats and each flat has a certain number of people living in it. The information about each building is given below.

Building A: The number of floors in building A is same as number of floors in building C. The number of flats per floor in is equal to the product of the smallest and the second smallest prime numbers. The total number of people living in building A is $\sqrt{1764}$.

Building B: The number of floors in building B is one less than the sum of flats per floor in building A and C. The number of flats per floor in building B is the same as the number of floors in building A. The total number of people in building B is $6\frac{2}{3}$ time the number of people in building A.

Building C: The number of floors in building is $\sqrt[4]{2401}$. The number of flats per floor in building C is one less than the number of flats per floor in building A. The number of people per flat in building C is four times more than the number of people per flat in building A. The total number of people in the building is 175.

Building D: The number of floors in building D is 9. The number of people per flat in building D is 60% of the number of people per flat in building C. The total number of people in building D is 108.

Note - The total number of people living in a particular building = No. of floors in the building x No. of flats on each floor x No. of people per flat.

Q16. What is the ratio of sum of total number of people living in building A and C to the sum of the people living in building B and D?

A. 217:398
B. 217:388
C. 219:388
D. 219:398
E. None of the above

Q17. The number of people living in building C is how much percentage less than the number of people living in building B?

A. 50%
B. 40%
C. 34%
D. 42.5%
E. 37.5%

Q18. Which among the four building has the highest number of flats and how many?

A. C, 35
B. A, 42
C. B, 70
D. C, 45
E. D, 36

Q19. If there is another build E where the number of floors is 70% of the number of floors in building B, the number of flats per floor is $\frac{1}{3}^{rd}$ more than the number of flats per floor in building A and the total number of people in the building is 280. Then find the number people per flat in building E.

A. 4
B. 2
C. 7
D. 3
E. 5

Q20. In the given options below the building and the corresponding number of people per floor is given. Find out which of the following option is right i.e., where the number of people per floor is correctly matched with the building name.

I. A – 6, D – 12, B – 28
II. B – 38, D – 12,
III. C – 25, A – 6, D – 12

A. Only I
B. Only III
C. Only I and III
D. Only II and III
E. Only I and II

Q21. Rs. 13000 was invested for 2 years in scheme A which offers compound interest, and the rate of interest ___% per annum. The amount received after 2 years from scheme A is Rs. 15,730. What approximate amount is received on investing the amount obtained from scheme A again in a different scheme B for 2 years where the interest rate of scheme B is twice the rate of interest of scheme A?

I. 10%, Rs 22,651

II. 12%, Rs 23,784

III. 15%, Rs 26,584

A. Only I
B. Only II
C. Only I and II
D. Only I and III
E. All I, II and III

Q22. Navya buys two articles A and B at the same cost price Rs. P. Then, she marks up both articles by 75% above their cost price. Then, she sold article A at Rs. 268 discount and article B at 20% discount. Then, which of the following statement(s) is/are definitely correct:

I: The profit earned by selling article B is greater than that by article A.

II: Discount percent given on article A is more than that in B

III: She earned equal profits by selling both the articles.

A. Only I
B. Only II
C. Both I and II
D. Only III
E. None of the above

Q23. Arun invested a certain sum of money at a rate of interest ___% for ___ years. If the ratio of the amount to interest is 216:91. Then find the rate of interest and time for which Arun invested the money.

I: 10%, 2 years

II: 20%, 3 years

III: 25%, 3 years

A. Only I
B. Only II
C. Both I and II
D. Both II and III
E. All I, II and III

Q24. A boat travels from town A to town B and then goes from town B to town C. Going from town A the boat reached B which is ___ km away in 5 hours while travelling downstream. Then the boats leave town B for C which is ___ km away from town B and reaches town C after 7 hours while travelling upstream. The ratio of the speed of boat to speed of stream is 8:1. (Note – Assume the speed of boat and stream is constant for both journeys)

I. 135 km, 147 km

II. 120 km, 147 km

III. 180 km, 196 km

A. Only II
B. Only III
C. Both I and II
D. Both I and III
E. Both II and III

Q25. Maya's monthly salary is 60% more than that of Swevi. Both Swevi and Maya, out of their respective monthly salary, pay equal sum towards EMI. Out of remaining monthly salary, Maya and Swevi, spend a certain amount towards house rent. Amount that Swevi pays towards EMI is 20% of her monthly salary. Amount that Maya pays towards house rent is 'x' times of that she pays towards EMI.

I: Find the total savings of Maya if her expenditure on EMI and house rent is just half of total salary. Also, EMI expense of Swevi is Rs 30000. It is to be assumed that Maya and Swevi had only two expenses from their salaries i.e., EMI and house rent.

II: Difference between house rents paid by Maya and Swevi is Rs 10000. House rent paid by Maya is Rs 6000 more than EMI paid by her. If house rent paid by Swevi is 33.33% of EMI paid by Maya, then, salary of Maya is?

A. Rs. 30000, Rs. 36000
B. Rs. 45000, Rs. 60000
C. Rs. 120000, Rs. 48000
D. Rs. 75000, Rs. 48000
E. Rs. 135000, Rs. 60000

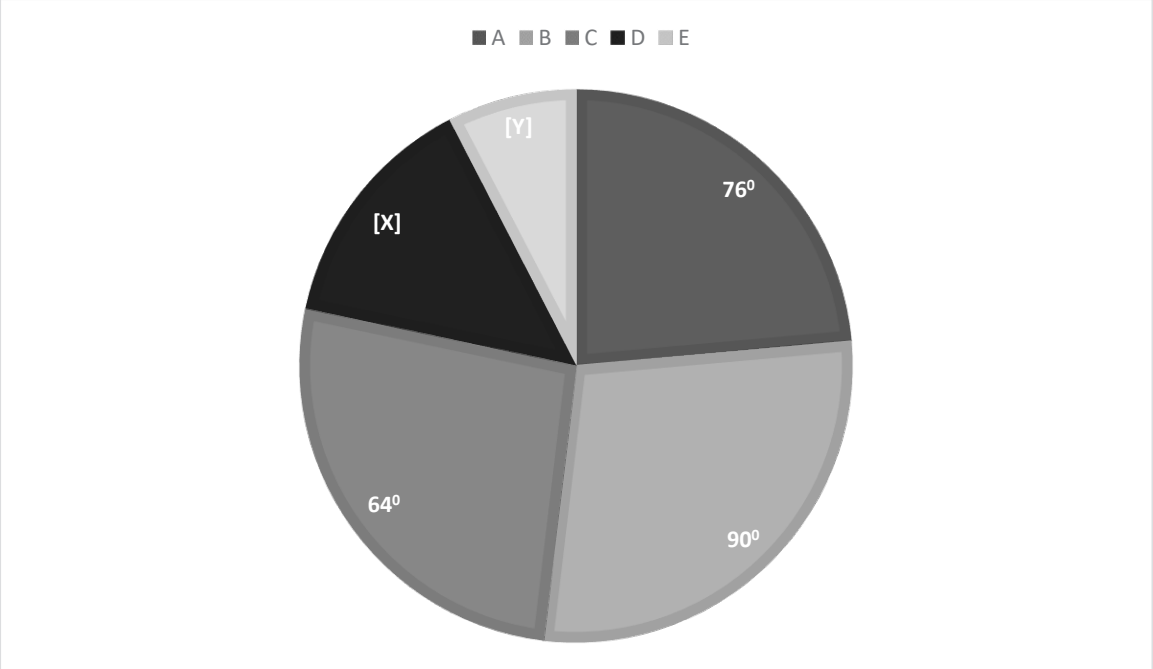
Direction (26-30): Five companies A, B, C, D and E manufacture bookshelves. Pie chart given below shows the distribution of total bookshelves manufactured by all 5 companies. The table here represents percentage distribution of unsold bookshelves of different companies.

Note:

1. The value of x is 26 degrees greater than that of B.

2. Bookshelves manufactured by D is 1160.

3. Total manufactured bookshelves = sold bookshelves + unsold bookshelves



COMPANY	Percentage Distribution
A	3P%
B	10%
C	4P%
D	20%
E	35%

Q26. Find the value of 11P – $\frac{3}{7}$y + $\frac{X}{29}$

A. 53
B. 51
C. 47
D. 59
E. 41

Q17. Find the difference between the unsold bookshelves of company B and average number of bookshelves sold by companies D and C.

A. 621
B. 595
C. 610
D. 630
E. None of the above

Q18. Find the difference between the average unsold bookshelves of company A and B and the average of sold bookshelves of company C and D.

A. 516
B. 618
C. 628
D. 544
E. None of the above

Q19. If there is another company F which manufactures 60% more bookshelves than company C and the number of unsold bookshelves for that company is 37.5% of the bookshelves manufactured by company F. Then find the number of bookshelves sold by company E?

A. 684
B. 712
C. 625
D. 610
E. 640

Q20. If half of the unsold bookshelves manufactured by all the companies are defective. Then find the ratio of sum of defective bookshelves from company B and D to sum of defective bookshelves from company A and C.

A. 161:121
B. 161:123
C. 173:121
D. 170:123
E. None of the above

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+91 8146207241 **hello@edutap.co.in** **edutap.in**